
UNIVERSITY OF SOUTH CAROLINA

HURRICANE RESCUE SOFTWARE

SOFTWARE REQUIREMENTS SPECIFICATION



CSCE247

The Fantastic 4

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1. INTRODUCTION

PURPOSE:

Hurricanes and tropical storms result in an average of 10,000 deaths per year. Hurricanes are low-pressure systems that form over warm ocean water and result in strong winds, tornadoes, floods, and heavy rain. These incidents cause many people to be stranded without essentials or medical care. Additionally, these storms can cause infrastructure damage, leaving many without homes or cars. Both category four and category five hurricanes are said to produce catastrophic damage due to intense wind speeds reaching at least 137 miles per hour, causing areas to become uninhabitable. The victims are left to start their lives over again, with little compensation, and hurricanes are only becoming more dangerous. With the increase in temperature due to climate change, there is warmer ocean water and moister air, resulting in stronger and more severe hurricanes. This is why we want to create a website that provides users with the ability to receive disaster relief in more locations and faster by ranking affected areas from most to least severely affected by the storms.

SCOPE:

This document will cover:

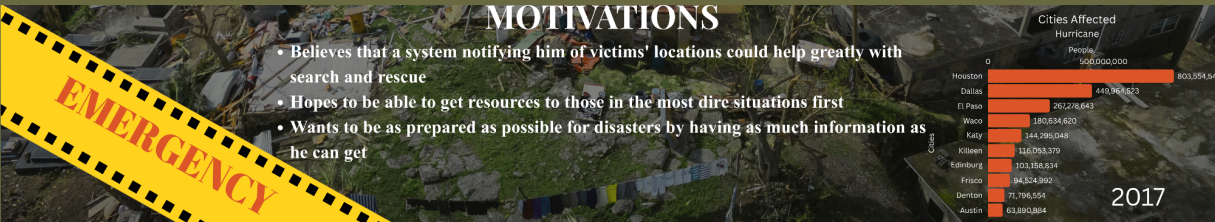
- The personas of potential users and stakeholders invested in this project
- Constraints implemented for this project
- Detailed description of the website
- Competitive Analysis to present and develop constraints and components for the website

2. STAKEHOLDERS

- **First responders** - professionals who are the first to arrive on the scene and aid victims
 - Paramedics
 - Firefighters
 - Law enforcement

- **Disaster relief groups** - organizations created to provide emergency aid, recovery support, and long-term assistance to communities affected by natural disasters
 - FEMA
 - United Way
 - American Red Cross
- **Victims of natural disasters** - individuals put in danger by the destruction caused by hurricanes
 - Family and friends of victims
- **Donors and volunteers** - individuals interested in contributing financially or through service to hurricane relief

3. PERSONAS

DARYL JONES	
	AGE: 42 OCCUPATION: Firefighter GENDER: Male LOCATION: Miami, Florida
BIO Daryl Jones is a firefighter who is specialized in hurricane and disaster rescue operations. He has experience with swiftwater and flood rescue, as well as search and rescue. His dad was a firefighter who raised him to value being there for others. Daryl suffered a hurricane at a young age and lost a close friend in a flood. This motivated him later in life to find a way he could help people with disaster relief. Seeing the destruction during this hurricane has led him to seek out better ways to reach those in need.	
GOALS	• Reaching people in need in a timely manner • Getting information on high-risk areas • Figuring out where to prioritize resource allocation
MOTIVATIONS • Believes that a system notifying him of victims' locations could help greatly with search and rescue • Hopes to be able to get resources to those in the most dire situations first • Wants to be as prepared as possible for disasters by having as much information as he can get	
	



Sissy Rosewood



AGE: 29

GENDER: Female

OCCUPATION: Actress

LOCATION: Los Angeles, California

BIO

Sissy Rosewood is a film actress based in Los Angeles, California. Sissy not only uses her time for acting in blockbuster movies, but also uses her time to help others as she prides herself as a humanitarian alongside an actress. She frequently donates to charities and support groups. Through social media, Sissy has learned about this tragic hurricane that has affected thousands of people in Texas. Devasted by the news, she has sought out to do as much as she can to help victims. With her huge platform that has garnered millions of followers and fans, she uses her stardom to spread awareness about the hurricane and sets up donations to victims in need of food, water, and shelter.

GOALS

- Spread awareness on social media to garner more traction for aid
- Donate millions to help with reconstruction
- Donates food, water, and provides resources for shelter

MOTIVATIONS

- Saving lives
- Rebuilding Communities
- Encourage charity work to those in need



Cities Affected Hurricane

Cities	People
Houston	803,554,542
Dallas	449,964,523
El Paso	267,278,643
Waco	180,634,620
Katy	144,295,048
Killeen	116,053,379
Edinburg	103,158,634
Frisco	94,524,992
Denton	71,796,554
Austin	63,890,984

2017

Claire Lunette



AGE: 56

GENDER: Female

OCCUPATION: Realtor

LOCATION: Houston, Texas

BIO

Claire is a realtor who sells homes in the Greater Houston area. She is hoping to retire in the next two years in order to spend more time with her son and his young family, but damages to her home during the hurricane may force her to abandon this plan. Given her occupation, a decrease in property values following the event may also affect her financial success in the short-term. She is concerned not only about her own well-being but also the safety of her daughter who lives in the city.

GOALS

- Locate friends and family during the event
- Getting access to basic resources (food, water, clothing, shelter)
- Verify when it is safe to return to her home

MOTIVATIONS

- Health and Safety
- Financial security
- Personal relationships



Cities Affected Hurricane

Cities	People
Houston	803,554,542
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2017

4. CONSTRAINTS

Time Constraints

- This project must be completed within 14 weeks
- Project members are full-time students with jobs

Monetary Constraints

- This project must be developed within a zero-dollar budget

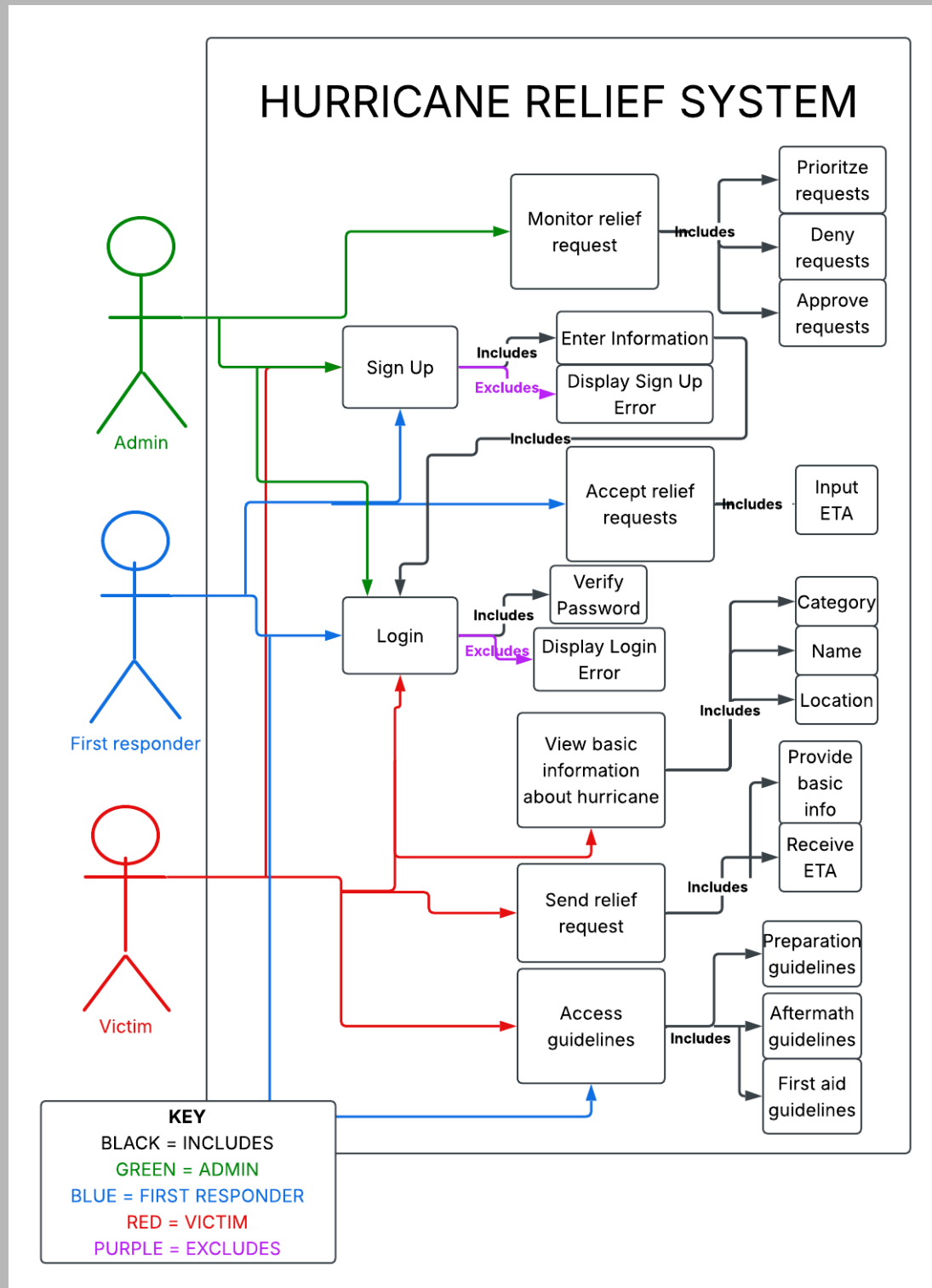
Technical Constraints

- The code must be written in Java
 - The website must be accessible through different device formats, including mobile devices, laptops, and PCs.
-

5. OVERALL DESCRIPTION

This app must be accessible to users in need during a hurricane crisis. Users will be able to access this app from mobile devices. The app will collect data on areas in need of disaster relief and send alerts to disaster relief groups, law enforcement, and first responders. Users affected by hurricanes will be able to send alerts and requests to the app that will be sorted from highest-risk to lowest-risk. The app will also include basic guidelines for first aid and hurricane response. Additionally, the app may need to be connected to text/call functions on the phone to allow easy communication with emergency services and loved ones.

5. BUSINESS USE CASES



6. FUNCTIONAL REQUIREMENTS

https://docs.google.com/spreadsheets/d/1PMYtaG7llvd-47X-E_QL47yH8hFZ1kAJzpQbitdH-tw/edit?usp=sharing

7. NON-FUNCTIONAL REQUIREMENTS

Look and Feel Requirements

- The user interface of the system shall be implemented in a way for beginner users to understand what they are looking at.
- Tabs and buttons in the user interface shall be large and easy to access in an emergency situation.

Usability Requirements

- The system shall be used by users with a wide range of capabilities.
- The system shall be used by users of a wide range of ages.

Performance Requirements

- The system shall be able to run on any device without an internet connection.

Maintainability and support requirements

- The system shall be able to run on Windows, Linux, and Mac systems.

Security requirements

- The user's information shall be stored in a secure manner.
- The user's information shall be stored only with their permission.

Legal Requirements

- The system shall comply with all state and federal law.
-

8. DEFINITIONS AND ACRONYMS

- **ALT text description:** A text explanation of graphics or images that can then be read aloud by a screen reader; for visually impaired users
 - **FEMA:** Federal Emergency Management Agency
 - **SAMHSA:** Substance Abuse and Mental Health Services Administration
 - **User interface:** Point of contact between the user and the system
 - **White labeling:** Taking a general product and adding another company's branding to it
-

9. COMPETITIVE ANALYSIS

Summary

After reviewing the apps described below, our group plans to follow the layout of the FEMA and Red Cross Emergency apps, but to make our app specifically designed for hurricanes. This will allow us to tailor our platform to a specific weather event, providing detailed information and safety tips for hurricanes that might be ignored on more general platforms. We also find the CrowdSource Rescue app very creative and effective in its use of civilian rescuers instead of solely relying on traditional systems. We plan to include some features inspired by the CrowdSource app by connecting volunteer rescuers with citizens. Researching accessibility features such as those provided on the FEMA app and including a Spanish-language option may also prove fruitful in making our product available to as many users as possible. Additionally, the FEMA and Red Cross apps are mainly intended for victims to use, but our app should include features that responders can use to help find and prioritize people in need.

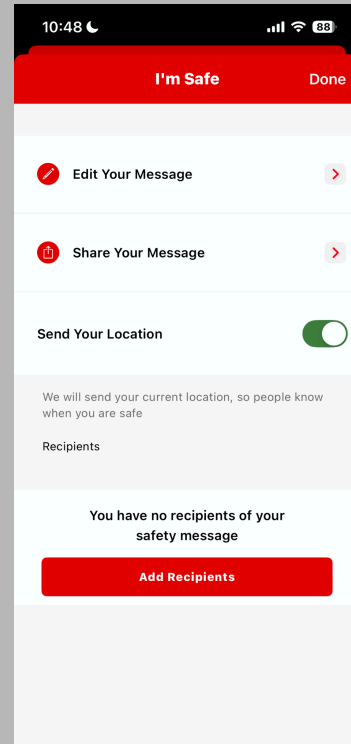
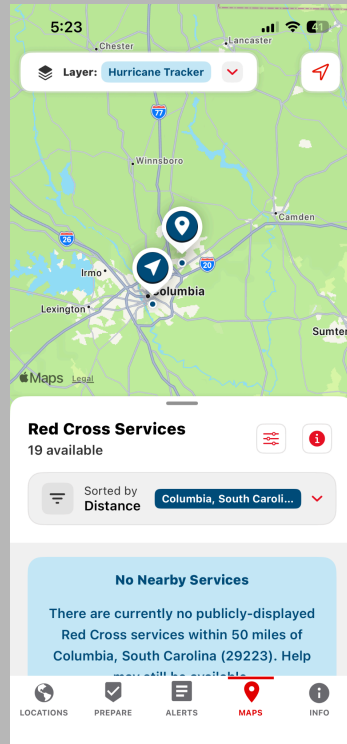
Table Summary

Name	Key components	Strengths	Weaknesses
Red Cross Emergency	FamilySafe feature Emergency alerts Shelter location map Safety guidelines and tips	+Offered in English and Spanish	-Not specific to hurricanes -Made primarily for victims
CrowdSource Rescue	Match victims with rescuers via GPS White labeling Early evacuation organization	+Tap into resources offered by civilian rescuers	-Not clear how civilian rescuers are vetted -Only available online
FEMA App	Weather and emergency alerts Map to locate shelters Disaster prep guides	+Accessible +English and Spanish	-Limited to the U.S. -Not specific to hurricanes

Red Cross Emergency App

Key features of the Red Cross Emergency App include a FamilySafe feature, emergency alerts, a shelter location map, and safety guidelines. Critically, the FamilySafe feature allows communication with family members who do not have the app. The message sent to family is set to “I’m safe” by default, allowing the user to avoid wasting time texting, but it can be edited if needed. Users can also send their location along with the message.

In addition to the FamilySafe feature, the app offers emergency alerts as well as a shelter location map. A weather tracker is integrated into the map, so users can view both shelters and weather conditions at once. In the Maps tab, there is also a link to call the Red Cross from the app. Overall, the app is well-organized and easy to navigate, with preparation guidelines, alerts, and maps organized into separate tabs. Our app should implement similar alerts and maps, but be tailored to hurricanes specifically, unlike the more general Red Cross Emergency app. We may also benefit from adding a feature similar to the FamilySafe tool.



CrowdSource Rescue Site

The CrowdSource Rescue site provides a unique way of organizing rescue efforts by emphasizing the role of civilians as rescuers. This can help take some of the burden off traditional first responders like firefighters who may be overwhelmed by the sheer volume of those in need. The site offers GPS tracking to match people with nearby civilians who can help rescue them. The civilian rescuers are supposedly vetted, although it is not explained how.

In addition to offering help during the disaster, CrowdSource coordinates with local churches to match volunteers with elderly citizens who need to evacuate before the event. The website also offers white labeling to integrate CrowdSource with existing volunteer groups. Ease of use and overall visual design are somewhat unclear since rescue features only open during a disaster. However, the CrowdSource Rescue promotional site, explaining the purpose of the app, is relatively clean and easy to read. Our app should allow mobile access, since web-only access limits who can use it. However, the concept of recruiting civilian volunteers is a useful one to implement in our own system since it utilizes often-overlooked resources.

WHAT IS CSR

CrowdSource Rescue is a Houston-based non-profit that uses next-generation technology to quickly connect both professional first-responders and vetted volunteers with response cases immediately before, during, and after a disaster.

We started in Harvey and have since operated in a number of disasters including Hurricane Michael, Hurricane Florence, Hurricane Maria, Hurricane Irma, Cyclone Maha, and a number of localized events.

RESPONSE MANAGEMENT

Using the latest GPS tracking and mapping technology, we connect our vetted and certified civilian rescuers with urgent cases before, during, and immediately after a disaster.

EARLY-EARLY RELIEF

Civilian volunteers are often the first, first responders. In the first 72 hours, they are first on scene with water, medical supplies, and other critical supplies while the logistic supply train gears up.

RAPID RESPONSE

During Harvey emergency services were overloaded with cases. We were able to quickly organize over 8,000 rescuers to assist their efforts in a matter of hours, allowing 911 to focus on critical cases.

EVACUATIONS

Before disasters the CSR platform is used by churches to connect elderly members with a vetted rescuer able to help them evacuate. The platform quickly identifies needs and matches them with volunteers.

RELIEF AND RECOVERY

We work with local municipalities and organizations to use our rescue data as a starting point for the long recovery process. We also gladly work with researchers, looking to learn from disasters and improve the process.

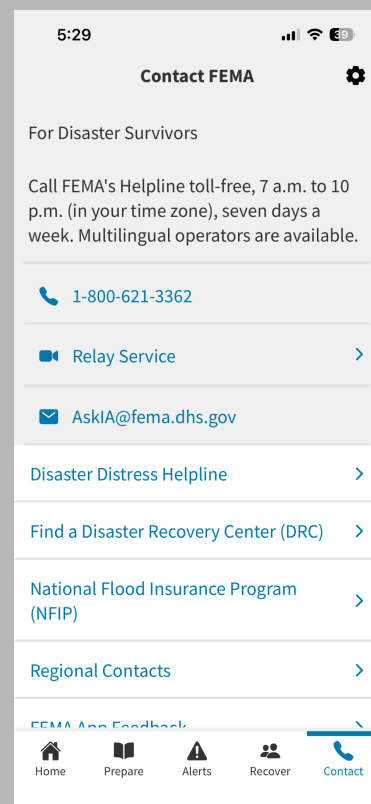
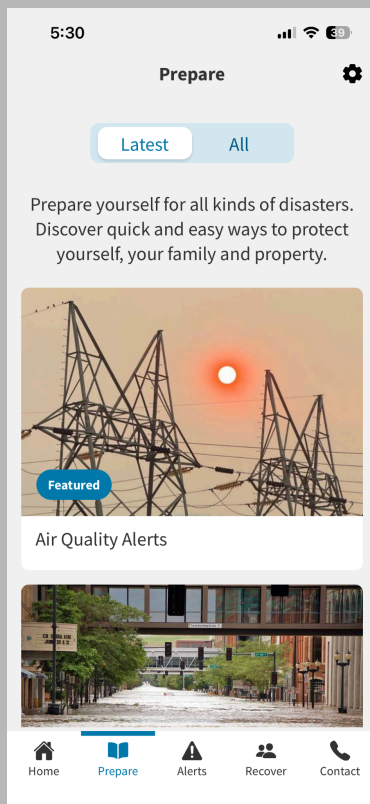
WHITE LABELING

Have a large existing volunteer base you want to mobilize in a disaster? Need to rapidly triage your employees after a disaster? CSR can offer a white-labeled version of the platform.

FEMA App

The Federal Emergency Management Agency app offers weather and hazard alerts for up to 5 locations specified by the user. Similar to the Red Cross Emergency App, the FEMA app provides a map to locate nearby Disaster Recovery Centers and shelters. Guidelines for how to prepare for disasters are also provided. Under the contact tab are links to call FEMA's Helpline as well as links to call or text a Disaster Distress Helpline organized by SAMHSA.

Because the app was made by the U.S. government, it follows federal accessibility requirements such as ALT text descriptions to make the app accessible to the visually impaired. However, the app is only available for use within the U.S. The app is overall very user-friendly, with an easy-to-find switch between Spanish and English, simple and visually appealing disaster preparation guides, and an FAQ menu. In order to remain competitive to a wide range of users, we should implement a Spanish language option as well as accessibility features. We will make our app available to users globally. (see screenshots on next page)



10. REFERENCES

<https://scied.ucar.edu/learning-zone/storms/hurricane-damage>

<https://www.nhc.noaa.gov/aboutsshws.php>

<https://www.redcross.org/about-us/news-and-events/news/Red-Cross-Has-an-App-for-That.html>

<https://crowdsourcerescue.org/>

<https://www.fema.gov/about/news-multimedia/mobile-products#faqs>